Higher National Diploma in Engineering (Mechanical/Electrical)
4th Year 1st semester Examination- 2018
ME 3227 – INDUSTRIAL ENGINEERING MANAGEMENT

Instructions for Candidates:	No. of questions	: 06
Answer any 5 questions.	No. of pages	: 04
	Time	: 3 hours

01)

i. What is meant by salvage value? (03 marks)

ii. There are two discounted cash flow techniques used to evaluate a project; that is NPV & IRR. Among these two, which one is the superior technique? Explain why it is superior. (03 marks)

iii. A businessmen is using the concept of NPV to assess two alternative methods for buying a packing machine which is expected to have a useful life time of 6 years.

Alternative I: Outright purchase for Rs.600,000.00, with a service contract of Rs.50,000.00 payable annually (at the beginning of each year) for 6 years. After 6 years the machine can be sold out for Rs.60,000.00

Alternative II: Lease a machine with an initial payment of Rs.200,000.00 and five further annual payments of Rs.100,000.00 inclusive of all service and maintenance charges. At the end, the machine can be sold out for Rs.60,000.00

a) Draw the cash flow diagrams separately for both alternatives. (04 marks)
b) For both alternatives calculate the NPV for a discount rate of 10%. (06 marks)
c) Explain, which alternative is more economical. (04 marks)

02).

i. Define what is meant by HR Management. (03 marks)

ii. Before more than 10-15 years the person who manage the human resource in a company is called Personal manager, later Human resource management has come in to act and way of managing human resource has been improved satisfactorily.

Discuss the benefits of a successful human resource management process. (06 marks)

iii. Explain the term Psychology and discuss the importance of Industrial Psychology. (06 marks)

iv. Explain the term motivation and discuss any motivation theory with reference to an industrial environment. (05 marks)

03).

i. What is meant by Financial Management? (03 marks)

ii. State 3 objectives of financial management. (06 marks)

iii. What is financial statement analysis? Describe in brief. (03 marks)

iv. Discuss the importance of financial statement analysis of a company considering their internal and external parties who get the use of that information. (08 marks)

04).

i. What is mean by "Quality"? (03 marks)

ii. Write short notes on following quality terms.

a. Quality Control
b. Quality Assurance (06 marks)

iii. Name any 3 product quality dimensions. (03 marks)

iv. State any 4 quality control tools and explain two of them. (08 marks)

05)

A)

i. Name any 3 types of organizational structures. (03 marks)

ii. Assume that you are going to start your own business. Select a suitable organizational structure for your intended business and discuss it under following topics.

a. What business you have selected
b. Organizational structure that you are going to use and the suitability of applying it for your organization.
c. Draw a suitable organizational chart (organigram).
d. Advantages and disadvantages of using the particular structure. (07 marks)

B)

A factory produces two types of polymeric items X and Y. The profit per unit of X is Rs. 7000.00 and profit per unit Y is Rs.8000.00.

Due to scarcity of skilled labour, the maximum number of X items that the factory can produce is limited to 4 units per day. A mixture of raw materials which cannot be used in the next day, is daily supplied by a subcontractor only if the orders are higher than 10 Kg. For production of each unit of X or Y, 2 Kg of that raw materials mixture is required.

Two curing plants with same facilities and each operating 24 hrs of the day, are available for daily production of X and Y. Product X is required to cure for 6 hrs and Product Y is required to cure for 8 hrs.

i) State the objective function and constrains. (04 marks)
ii) Using graphical method determine the numbers of X and Y to be produced in order to maximize the profit. (06 marks)

06)

i. List any 3 types of facility layouts. (02 marks)

ii. Write short notes on any 2 of above facility layouts. (06 marks)

iii. The procedure table for activities required to produce an electrical item is given below. The plant operates 12 hrs per day. Daily demand for the products is 48 items.

Activity	Predecessor	Time (minutes)
A	-	3
B	-	4
C	B	4
D	A,C	5
E	A	2
F	E	4
G	E	6
H	G	8
I	D,F	6
J	G,I	14
K	J,H	12
L	K	12

a) Draw the network diagram. (04 marks)

b) Calculate the cycle time. (02 marks)

c) Assign the tasks in to workstations, considering a minimum possible number of workstations. (04 marks)

d) Calculate efficiency of the line. (02 marks)